

ORIGINAL ARTICLE

Early DOCA-Salt Hypertension Develops Without Increases in Renal or Lumbar Sympathetic Nerve Activity

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BACKGROUND: Excessive salt intake is a major contributor to hypertension; however, the role of sympathetic nerve activity (SNA) remains controversial. Most existing evidence relies on indirect markers or single-time-point measurements. Therefore, we directly and continuously recorded SNA to test the hypothesis that SNA contributes to the initiation, progression, or both phases of salt-induced hypertension.

METHODS: We continuously recorded renal and lumbar SNA, arterial pressure, and heart rate for 4 weeks in conscious sham-operated and uninephrectomized rats. After a control period on a normal-salt diet, rats received a subcutaneous deoxycorticosterone acetate (DOCA) pellet with a high-salt diet (4% NaCl) for 17 days, followed by pellet removal and a return to a normal-salt diet for 7 days.

RESULTS: DOCA-salt treatment induced a biphasic increase in arterial pressure, with an immediate increase within 24 hours (initiation phase) and a progressive increase after day 6 (developmental phase), which was more pronounced in uninephrectomy rats. In contrast, heart rate and renal SNA decreased from day 1, and lumbar SNA progressively declined in all groups, providing no evidence of sympathetic overactivity during either phase. After DOCA withdrawal, AP declined, whereas renal SNA and lumbar SNA transiently increased.

CONCLUSIONS: Continuous recordings demonstrated that neither renal SNA nor lumbar SNA mediated the initiation or development of DOCA-salt hypertension. In the absence of evidence for increased cardiac output, the biphasic rise in AP is attributed to increased peripheral vascular resistance of a nonsympathetic origin, likely driven by DOCA-sodium interactions in the perivascular region. (*Hypertension*. 2026;83:2731–2742. DOI: 10.1161/HYPERTENSIONAHA.125.26423.) • **Supplement Material.**

Key Words: acetate ■ arterial pressure ■ cardiac output ■ diet ■ heart rate

Excessive dietary salt intake is a well-established contributor to the onset and progression of hypertension. Sympathetic nerve activity (SNA) has frequently been reported to increase in hypertensive states,^{1,2} suggesting a potential causal link among high salt intake, increased SNA, and the development of hypertension. In contrast, salt-induced hypertension has also been described as involving complex, multifactorial processes that act in a time-dependent and interactive manner.^{3,4} Consequently, it remains unclear whether augmented SNA serves as a primary driver that directly increases arterial pressure

(AP) or whether it represents a secondary response to established hypertension. Distinguishing causal mechanisms from associative responses is critical, yet current evidence remains inconclusive.⁴

The deoxycorticosterone acetate (DOCA)-salt model of hypertension is widely used to elucidate the pathogenesis of salt-induced hypertension because it allows for clear delineation of initiation, development, and maintenance phases.^{3,4} A longstanding hypothesis suggests that sympathetic overactivity plays a crucial role in the initiation and progression of DOCA-salt hypertension.

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Novelty and Relevance

What Is New?

This study provides the first long-term, continuous recordings of renal and lumbar sympathetic nerve activities (RSNA and LSNA) in conscious rats during the initiation and early development of DOCA-salt hypertension. Contrary to the prevailing view, both RSNA and LSNA declined even as arterial pressure (AP) progressively increased.

What Is Relevant?

These findings challenge the longstanding hypothesis that sympathetic overactivity drives the onset of salt-sensitive hypertension. Instead, they suggested that increased peripheral vascular resistance, mediated by nonsympathetic mechanisms, underlies the initiation and early development of DOCA-salt hypertension.

Clinical/Pathophysiological Implications?

Therapeutic strategies directed at suppressing sympathetic activity, including renal denervation, are unlikely to be effective in the early phase of salt-induced hypertension. Rather, these findings point to the potential efficacy of interventions targeting vascular dysfunction, possibly through modulation of immune-inflammatory pathways.

Nonstandard Abbreviations and Acronyms

Ang II	angiotensin II
AP	arterial pressure
DOCA	deoxycorticosterone acetate
EC-MR	endothelial cell mineralocorticoid receptor
HR	heart rate
LSNA	lumbar sympathetic nerve activity
MAP	mean arterial pressure
NREM	nonrapid eye movement
RSNA	renal sympathetic nerve activity
SNA	sympathetic nerve activity

AP, RSNA, and LSNA from a normotensive baseline through the initiation and development of hypertension, we aimed to elucidate the temporal relationships among these variables. This approach enables rigorous testing of whether time-dependent changes in SNA contribute to the initiation and progression of hypertension, or instead represent secondary responses to other underlying mechanisms.

METHODS

Data Availability

Detailed Methods are provided in the [Supplemental Material](#). The data sets supporting the present findings are available from the corresponding author on request.

Animals

Male Wistar rats were used in all the experiments. Animals were housed individually in temperature-controlled chambers (24 °C; Espec, Osaka, Japan) under a 12:12-hour light–dark cycle. Food and water were provided ad libitum.

Ethical Considerations

All experimental procedures conformed to the Guiding Principles for the Care and Use of Animals in the Field of Physiological Sciences (Physiological Society of Japan, 2002)¹² and were approved by the Animal Care and Use Committee of Nara Women’s University.

Surgical Procedures

Right Nephrectomy

Two weeks before the start of the experiment, rats were anesthetized with isoflurane (1.5%–4%) following premedication with sodium pentobarbital (65 mg/kg, IP). A flank incision was

However, much of the supporting evidence derives from indirect indicators, such as plasma norepinephrine levels⁵ or findings from sympathetic denervation studies,^{6,7} and conflicting reports continue to fuel the debate. Some studies report that increased sympathetic activity is a direct cause of hypertension,^{1,8} whereas others argue against its involvement.^{5,9} A major limitation underlying this controversy is the scarcity of longitudinal data directly measuring SNA during the transition from normotension to hypertension.⁴ Indirect measurements, typically obtained at discrete time points, offer only a transient snapshot of SNA, making it difficult to determine whether changes in SNA precede, follow, or coincide with increases in AP.

In the present study, we sought to address this issue by continuously recording renal SNA (RSNA) and lumbar SNA (LSNA) throughout the course of DOCA-salt treatment in the same animals, using previously established methods.^{10,11} By capturing dynamic changes in

made to expose the right kidney, and the adrenal gland was separated from the upper pole. The kidney was decapsulated, and the renal artery, vein, and ureter were ligated together as a bundle using 4-0 silk suture before kidney removal. For sham nephrectomy (sham) procedures, the right kidney was exposed, and the perirenal fat was gently manipulated without kidney removal, after which the incision was closed.

Surgical Instrumentation

One week before the start of the experiment, rats were anesthetized as described above. Electrodes for recording RSNA and LSNA, and a telemetry device for measuring AP (TA11PA-C40; DSI, New Brighton, MN) were chronically implanted. Detailed procedures have been described previously^{10,11,13} and are also provided in the [Supplemental Material](#).^{14,15}

DOCA Silicone Implantation

DOCA-silicone pellets were prepared by mixing DOCA (100 mg; FUJIFILM, Tokyo, Japan) with 2 mL of a silicone elastomer base (Sylgard 184; Dow Corning, Midland, MI), following the method described by Jacob et al.⁶ After a 4-day control period, rats were lightly anesthetized with isoflurane, and each DOCA-silicone pellet was cut into 2 to 3 mm cubes (equivalent to 100 mg DOCA per rat) and placed subcutaneously between the scapular blades. After 17 days of DOCA treatment combined with a high-salt diet (4% NaCl), rats were again lightly anesthetized, and the DOCA-silicone pellets were removed.

Experimental Protocols

Recordings were performed in an electronically shielded, sound-attenuated chamber (Espec, Osaka, Japan) maintained at a constant temperature (24 °C) under a fixed light-dark cycle (lights on from 5:00 AM to 5:00 PM). At baseline, body weights were 249.8±3.0 g (n=12) in the sham group, 253.3±3.8 g (n=14) in the Sham-DOCA-salt group, and 255.5±3.4 g (n=14) in the uninephrectomy-DOCA-salt group.

Rats in the sham group were maintained on a normal-salt diet (0.535% NaCl; Oriental Yeast, Osaka, Japan) throughout the experimental period. Based on food intake and body weight during the control phase, sodium intake was estimated to be ≈1.83 mmol/d (≈7.0 mmol/kg per day). In the sham-DOCA-salt and uninephrectomy-DOCA-salt groups, all rats were initially maintained on the same normal-salt diet for a 4-day control period. Under light isoflurane anesthesia (1.5% to 4%), a DOCA pellet was implanted subcutaneously, after which the diet was switched to a high-salt regimen (4% NaCl; Oriental Yeast, Osaka, Japan) for 17 days. Subsequently, under identical anesthetic conditions, the DOCA pellet was removed, and the rats were returned to the normal-salt diet for a 7-day recovery period.

The implantation and removal surgeries were performed during the final hour of the control and DOCA-salt treatment periods, each lasting ≈15 minutes. Sham-operated rats underwent identical sham implantation and removal procedures and were maintained on a normal-salt diet throughout the experimental period.

Throughout the 24-day experimental period, all physiological variables were continuously recorded for 24 h/d, with a ≈20-minute daily interruption for body weight measurement, animal care, and data processing.

The animals were visually monitored by video camera for 90 minutes every other day during the light phase (09:30–11:00). Behavioral states were classified as nonrapid eye movement (NREM) sleep, quiet wakefulness, moving, grooming, eating, or drinking. Behavioral scoring was performed on a second-by-second basis by an investigator with extensive experience in rat behavioral assessment, based on real-time visual inspection of the video recordings during the experiments, followed by posthoc identification of rapid eye movement sleep periods.¹⁶ NREM sleep was defined as the visually observed sleep state, excluding periods estimated as rapid eye movement sleep, which were identified based on characteristic heart rate (HR) changes.

RSNA and LSNA waveforms and sound signals were monitored daily. Recordings in which neural activity patterns were markedly suppressed and disappeared for several consecutive days were considered unsuccessful and excluded from further analysis.

Data Acquisition

Details are provided in the [Supplemental Material](#).

Data Analysis

Percentage changes in RSNA and LSNA were calculated using the mean values recorded during the light (daytime) period of the 3-day control period as the 100% reference. Detailed descriptions of the quantification procedures for RSNA and LSNA, including signal-to-noise characteristics and the stability of long-term recordings, are provided in the [Supplemental Material](#). Baseline levels of mean arterial pressure (MAP), HR, RSNA, and LSNA were determined as the average values obtained during NREM sleep.^{13,15,17} The diurnal amplitudes of MAP, HR, RSNA, and LSNA were defined as the differences between the daily peak and daily nadir values.

Statistical Analysis

Data are presented as mean±SEM. Time-course data were analyzed using 2-way repeated-measures ANOVA, with time as the within-subject factor and group as the between-subject factor. When significant main effects or interactions were detected, posthoc analyses were performed. Within-group comparisons versus the control period were conducted using Dunnett test, whereas between-group comparisons at corresponding time points were performed using Holm-Bonferroni-adjusted pairwise tests. Adjusted *P* values are reported throughout, and *P* < 0.05 was considered statistically significant. All analyses were performed using R (version 4.5.3; R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

Time Course of Changes in AP, HR, RSNA, and LSNA During the Initiation and Development of Hypertension

Figure 1 presents representative recordings of AP, RSNA, and LSNA obtained from an uninephrectomy rat during the control period (day 4), the DOCA-salt

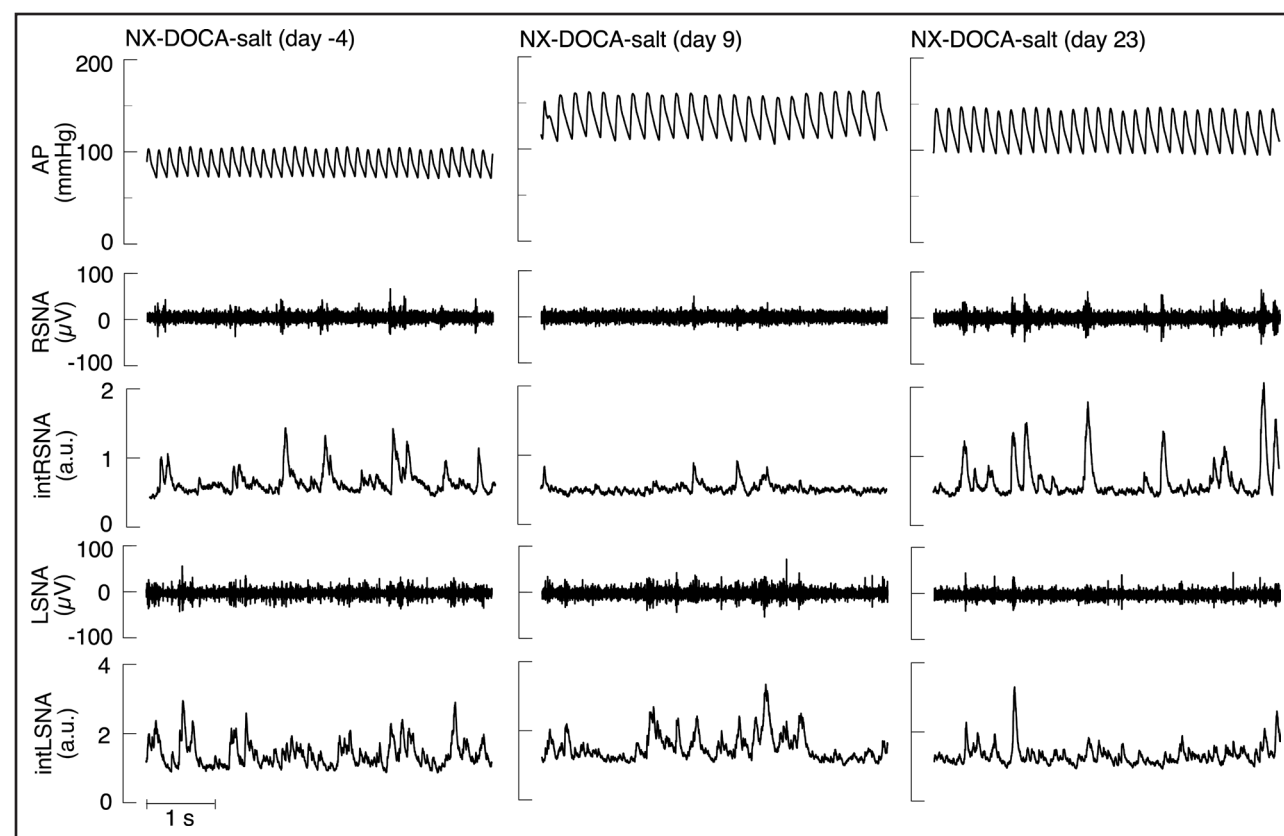


Figure 1. Representative recordings from an individual uninephrectomized (NX) rat.

Traces show arterial pressure (AP), renal sympathetic nerve activity (RSNA), integrated RSNA (intRSNA), lumbar sympathetic nerve activity (LSNA), and integrated LSNA (intLSNA) obtained during the sleeping period. Data were continuously recorded throughout the experiment, and representative segments were selected from 3 phases: the control period (day 4), the deoxycorticosterone acetate (DOCA)–salt treatment period (day 9), and the recovery period after DOCA–salt treatment (day 23). Integrated RSNA and LSNA are expressed in arbitrary units

treatment phase (day 9), and the recovery period following DOCA–salt treatment (day 23). AP was elevated during the high-salt plus DOCA phase, accompanied by a reduction in RSNA, whereas LSNA showed little apparent change. Following the removal of the DOCA pellet and the return to a normal-salt diet, AP remained elevated relative to the control period, whereas both RSNA and LSNA returned to levels comparable to those observed during the control period. Thus, representative short-term recordings of AP changes induced by uninephrectomy–DOCA–salt loading did not reveal clear temporal changes in RSNA or LSNA that were directly associated with the elevation in AP. [Figure S2](#) presents beat-by-beat cardiac cycle-triggered overlay plots of AP, RSNA, and LSNA to further illustrate the short-term temporal relationships among these variables.

Figure 2 shows the time course of hourly mean values for MAP, HR, RSNA, and LSNA in the sham–DOCA–salt and uninephrectomy–DOCA–salt groups across 3 phases: a 4-day control, a 17-day DOCA plus high-salt treatment, and a 7-day recovery after DOCA removal. Rats in the sham group were maintained on a normal-salt diet throughout the experiment. These continuous

4-week recordings showed clear time-dependent trends while preserving diurnal variations.

MAP in the uninephrectomy–DOCA–salt group demonstrated a pronounced biphasic increase. The initial phase was characterized by a rapid rise in MAP, which occurred similarly in both the sham–DOCA–salt and uninephrectomy–DOCA–salt groups, followed by a more gradual rise through day 6. Thereafter, MAP continued to increase more prominently in the uninephrectomy–DOCA–salt group, resulting in significant differences compared with the sham–DOCA–salt group between days 5 and 17 (indicated by no. 23 in Figures 2 and 3). After removal of DOCA and the return to a normal-salt diet, MAP declined immediately and then progressively declined from day 18 to day 24 in both DOCA-treated groups.

HR and RSNA were significantly reduced by DOCA–salt treatment in both the sham–DOCA–salt and uninephrectomy–DOCA–salt groups, with no additional effect of uninephrectomy on their time courses. LSNA declined progressively over time in all groups (sham, sham–DOCA–salt, and uninephrectomy–DOCA–salt), indicating no influence of either DOCA–salt treatment or uninephrectomy. To characterize overall patterns, continuous

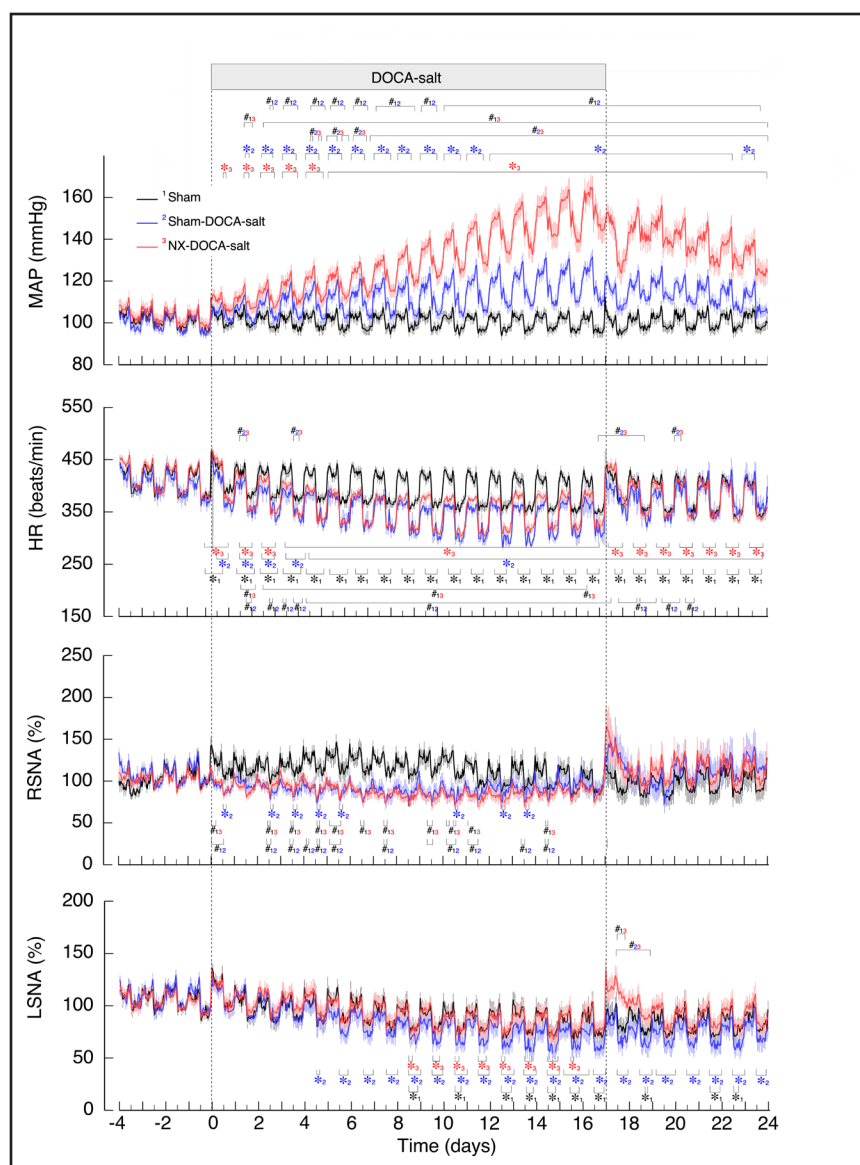


Figure 2. Time course of arterial pressure (AP), heart rate (HR), integrated renal sympathetic nerve activity (RSNA), and integrated lumbar sympathetic nerve activity (LSNA) in the Sham ($n=12, 12, 10, 10$), Sham–deoxycorticosterone acetate (DOCA)–salt ($n=14, 14, 9, 10$), and uninephrectomy (NX)–DOCA–salt ($n=14, 14, 11, 9$) groups, respectively.

Solid lines represent group means, and thin vertical bars denote $\pm$ SE. Black, blue, and red lines indicate the Sham, Sham–DOCA–salt, and NX–DOCA–salt groups, respectively. Asterisks indicate significant differences ($P<0.05$) from the corresponding control-period mean within the same group during the light (day) or dark (night) phases. The color of each asterisk corresponds to the experimental group: black for Sham (*1), blue for Sham–DOCA–salt (*2), and red for NX–DOCA–salt (*3). Pound symbols (#) denote significant pairwise differences ($P<0.05$) between groups at corresponding time points: no. 12, Sham vs Sham–DOCA–salt; no. 13, Sham vs NX–DOCA–salt; and no. 23, Sham–DOCA–salt vs NX–DOCA–salt. The duration of statistical significance is indicated by thin horizontal lines, with the asterisks or pound symbols above these lines identifying the corresponding comparisons. Symbols located above the mean values indicate significant increases, whereas symbols located below the mean values indicate significant decreases relative to the corresponding reference condition.

data were summarized in terms of basal levels and diurnal variations, thereby consolidating the data set and quantifying time-dependent changes.

Figure 3A shows the changes in basal MAP, HR, RSNA, and LSNA during NREM sleep. DOCA-salt treatment elicited an immediate and significant elevation in MAP on day 1 by 6.9 ± 0.6 mmHg ($P=0.048$ [95% CI, 0.0–13.6 mmHg]) and 6.8 ± 0.9 mmHg ($P=0.046$ [95% CI, 0.1–13.8 mmHg]) in sham–DOCA-salt and uninephrectomy–DOCA-salt groups, respectively. In the uninephrectomy–DOCA-salt group, MAP continued to rise significantly from day 7, reaching 40.4 ± 1.8 mmHg ($P<0.001$ [95% CI, 28.3–52.5 mmHg]) on day 17, which exceeded the sham–DOCA-salt group by 27.7 ± 1.4 mmHg ($P<0.001$ [95% CI, 13.1–42.3 mmHg]). After withdrawal of DOCA-salt treatment, MAP progressively declined toward baseline in both DOCA-treated groups.

HR decreased from day 1 by -22.2 ± 2.2 bpm ($P=0.027$ [95% CI -42.7 to -1.7 bpm]) and -26.2 ± 2.1 bpm ($P=0.006$ [95% CI -46.7 to -5.7 bpm]) in the sham–DOCA-salt and uninephrectomy–DOCA-salt groups, respectively, and remained suppressed throughout the DOCA-salt treatment period. Although a similar trend was observed in the sham group, the magnitude of HR reduction was significantly greater in both DOCA-treated groups. After termination of DOCA-salt treatment, HR during NREM sleep gradually returned to baseline levels in both the sham–DOCA-salt and uninephrectomy–DOCA-salt groups.

RSNA decreased significantly from day 1 in the uninephrectomy–DOCA-salt group ($-24.4\pm 7.7\%$; $P=0.011$ [95% CI, -44.0% to -4.8%]) and from day 3 in the sham–DOCA-salt group ($-20.8\pm 8.5\%$; $P=0.043$ [95% CI, -42.6% to -1.0%]) relative to the sham group, and remained suppressed until day 11. From days 15 to 17,

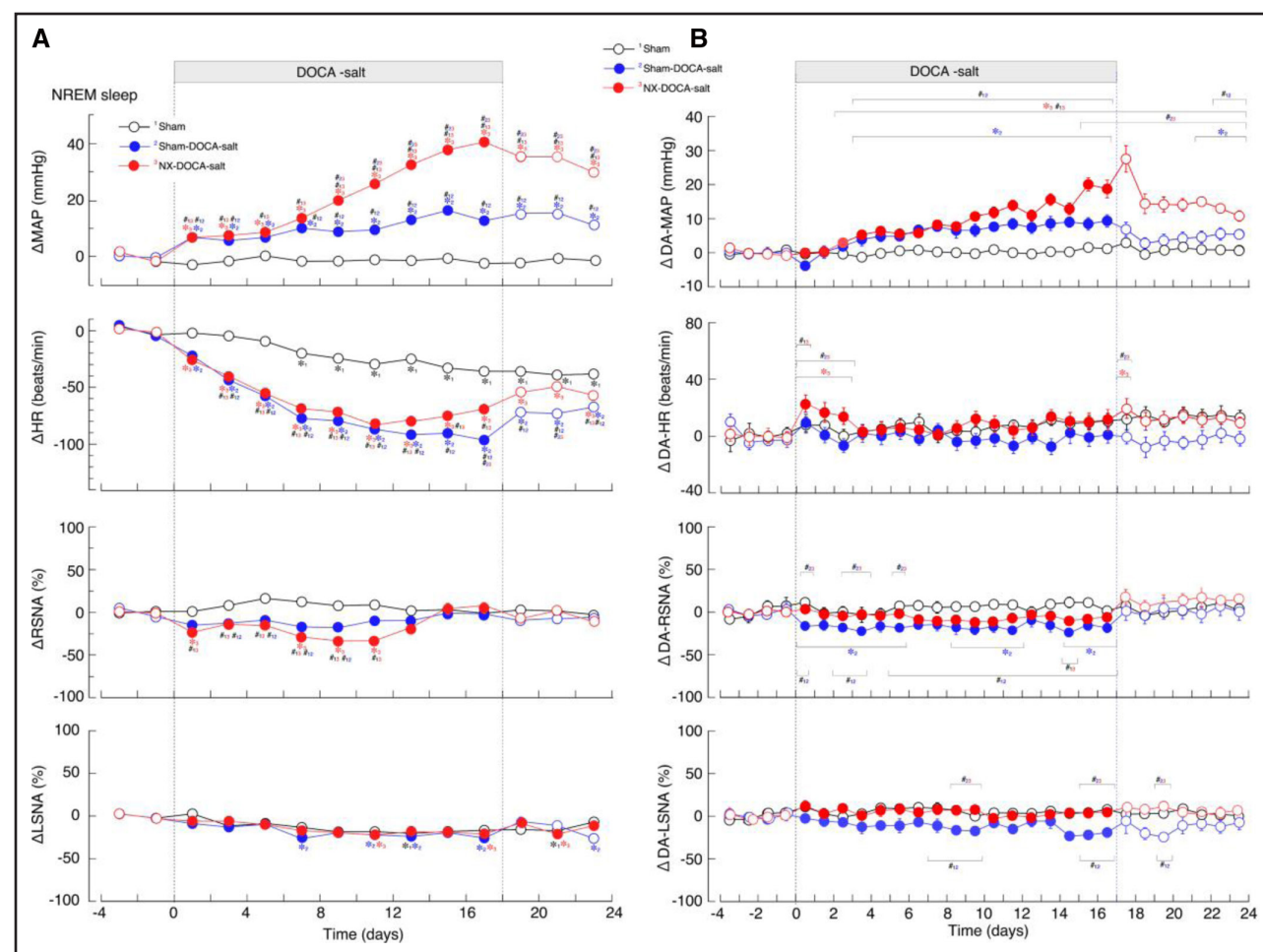


Figure 3. Time course of changes in mean arterial pressure (MAP), heart rate (HR), renal sympathetic nerve activity (RSNA), and lumbar sympathetic nerve activity (LSNA), expressed as Δ values relative to the control period.

A, Changes in MAP, HR, RSNA, and LSNA during non-rapid eye movement (NREM) sleep. Values obtained during NREM sleep were considered to represent basal levels. **B, Diurnal amplitude (DA),** defined as the difference between values during the active (dark) and inactive (light) phases, for MAP, HR, RSNA, and LSNA, also expressed as Δ values from the control period. Symbols and statistical annotations are defined in the legend to Figure 2. DOCA indicates deoxycorticosterone acetate.

RSNA gradually increased and returned to control levels in both groups. After cessation of DOCA-salt treatment, RSNA during daytime NREM sleep did not differ significantly from the respective control-period levels in any group (all $P>0.05$) or among groups at corresponding time points (all $P>0.05$).

LSNA showed intermittent significant decreases ($P<0.05$) in the sham-DOCA-salt group on days 7, 13, and 17; in the uninephrectomy-DOCA-salt group on days 11 and 17; and in the sham group on day 13. After cessation of DOCA-salt treatment, LSNA during daytime NREM sleep showed sporadic decreases below control levels. Significant reductions were observed on day 21 in the sham group ($-17.1\pm5.2\%$; $P=0.035$ [95% CI, -35.5% to -0.8%]) and the uninephrectomy-DOCA-salt group ($-21.2\pm5.7\%$; $P=0.010$ [95% CI, -38.4% to -4.0%]), and on day 23 in the sham-DOCA-salt group ($-26.2\pm6.8\%$; $P=0.007$ [95% CI, -46.7% to -5.6%]).

Figure 3B shows changes in the diurnal amplitudes of MAP, HR, RSNA, and LSNA. DOCA-salt treatment

increased the diurnal MAP amplitude from day 3 in the uninephrectomy-DOCA-salt group to day 4 in the sham-DOCA-salt group. In the uninephrectomy-DOCA-salt group, diurnal MAP amplitude further increased from day 10, reaching 18.8 ± 2.6 mmHg ($P<0.001$ [95% CI, 13.1 – 24.6 mmHg]) on day 17, whereas in the sham-DOCA-salt group, it remained elevated at a plateau level of 9.3 ± 1.8 mmHg ($P<0.001$ [95% CI, 3.6 – 15.1 mmHg]) on day 17. The between-group difference on day 17 was 9.4 ± 2.6 mmHg ($P=0.001$ [95% CI, 3.1 – 15.9 mmHg]). Following the removal of the DOCA pellet and reinstatement of the normal-salt diet, the diurnal amplitude of MAP decreased rapidly. Between days 16 and 24, diurnal MAP amplitude in the uninephrectomy-DOCA-salt group was significantly higher than that in the sham-DOCA-salt group.

The diurnal amplitude of HR transiently increased on days 1 to 2 in the uninephrectomy-DOCA-salt group but showed no further changes, and remained unaltered in the sham-DOCA-salt and sham groups.

RSNA amplitude was intermittently reduced in the sham–DOCA-salt group compared with control, with no consistent changes in the sham or uninephrectomy–DOCA-salt groups. LSNA amplitude remained unchanged in all groups, except for sporadic short periods of intergroup differences.

Water Balance and Body Weight

As shown in Figure 4, water intake and urine volume increased stepwise immediately after the initiation of DOCA-salt treatment in both sham–DOCA-salt and uninephrectomy–DOCA-salt groups. These parameters

remained elevated throughout the treatment period and returned to baseline after cessation. The overall time courses did not differ significantly between the 2 groups.

In the uninephrectomy–DOCA-salt group, water balance increased significantly ($P<0.05$) on days 3 and 17 during DOCA-salt treatment compared with the control level. However, water balance during DOCA-salt treatment did not differ significantly from that in the sham–DOCA-salt group. Both DOCA-treated groups exhibited intermittent increases compared with the sham group, although these differences were not sustained over consecutive days. Body weight increased linearly

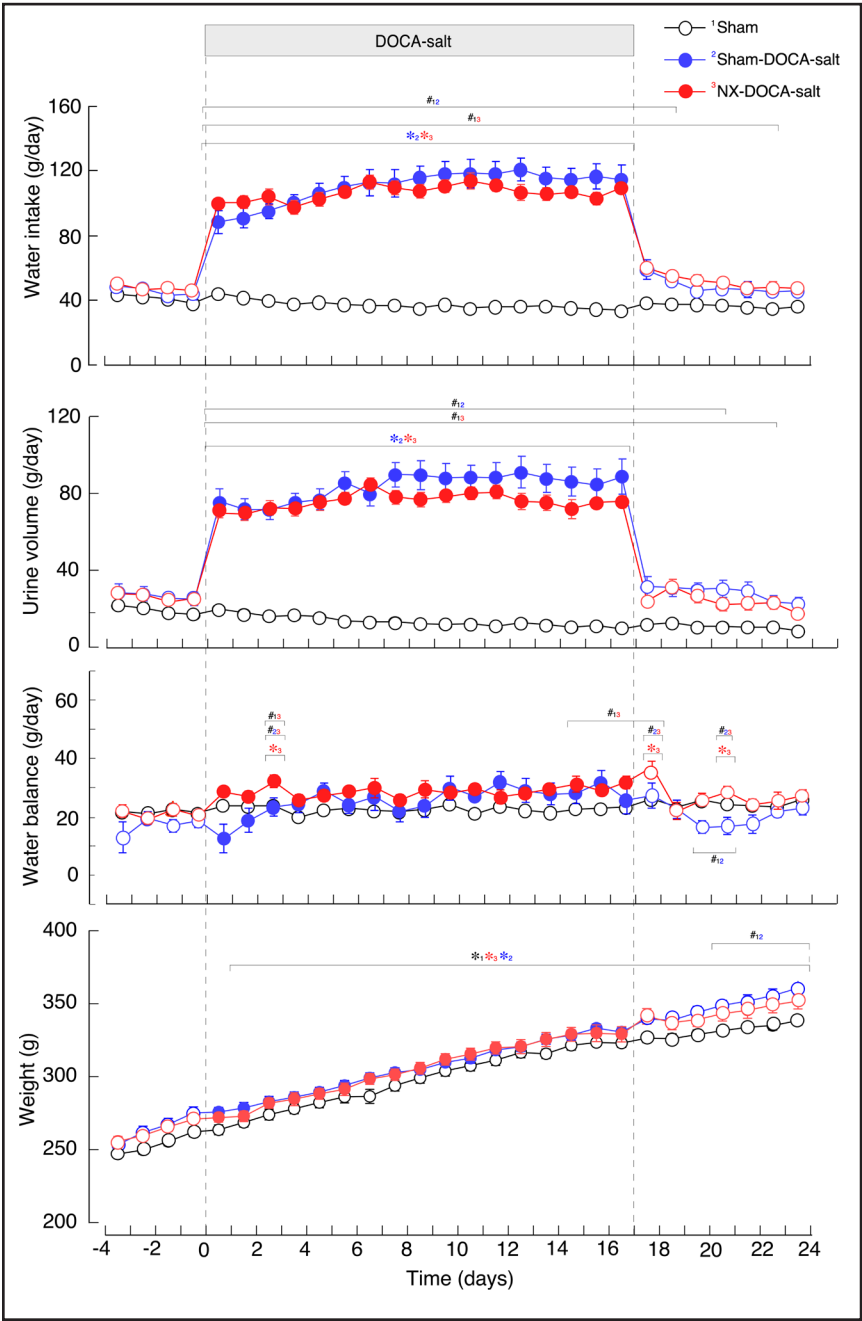


Figure 4. Time course of daily water intake, urine volume, water balance, and body weight in the Sham, Sham–deoxycorticosterone acetate (DOCA)–salt, and uninephrectomy (NX)–DOCA–salt groups. Each parameter represents the total value measured over a 24-hour period. Symbols and statistical annotations are defined in the legend to Figure 2.

throughout the experimental period in all 3 groups, with no significant differences among groups during DOCA-salt treatment.

Behavioral Patterns

Figure 5 shows the time course of behavioral patterns recorded every other day during the light period across the control, DOCA-salt, and recovery phases. Within each group, prolonged sleep duration and fluctuations in moving time relative to baseline were observed sporadically. Nevertheless, no consistent or sustained differences were detected. Statistical analysis revealed no significant differences among the 3 groups in the occurrence of each behavioral state.

DISCUSSION

This study aimed to continuously and simultaneously record RSNA and LSNA in uninephrectomized

(uninephrectomy) and sham-operated rats for 4 weeks before, during, and after DOCA-salt treatment. This longitudinal design enabled quantitative assessment of the magnitude and time course of RSNA and LSNA changes in response to a high-salt diet combined with DOCA, and to examine how uninephrectomy influences the initiation and progression of hypertension.

We demonstrated that DOCA-salt treatment in the uninephrectomy-DOCA-salt rats produced 2 distinct phases of AP elevation: an initiation phase (days 1–5), characterized by a rapid increase within 24 hours of treatment, and a developmental phase (days 6–17). These findings are consistent with previous reports.^{3,4} After the cessation of DOCA-salt treatment, AP decreased, in agreement with earlier studies.¹⁸ Simultaneous measurements of RSNA, LSNA, and HR provided direct insight into whether these variables contribute to the 3 characteristic AP responses induced by DOCA-salt treatment: initiation, development, and recovery.

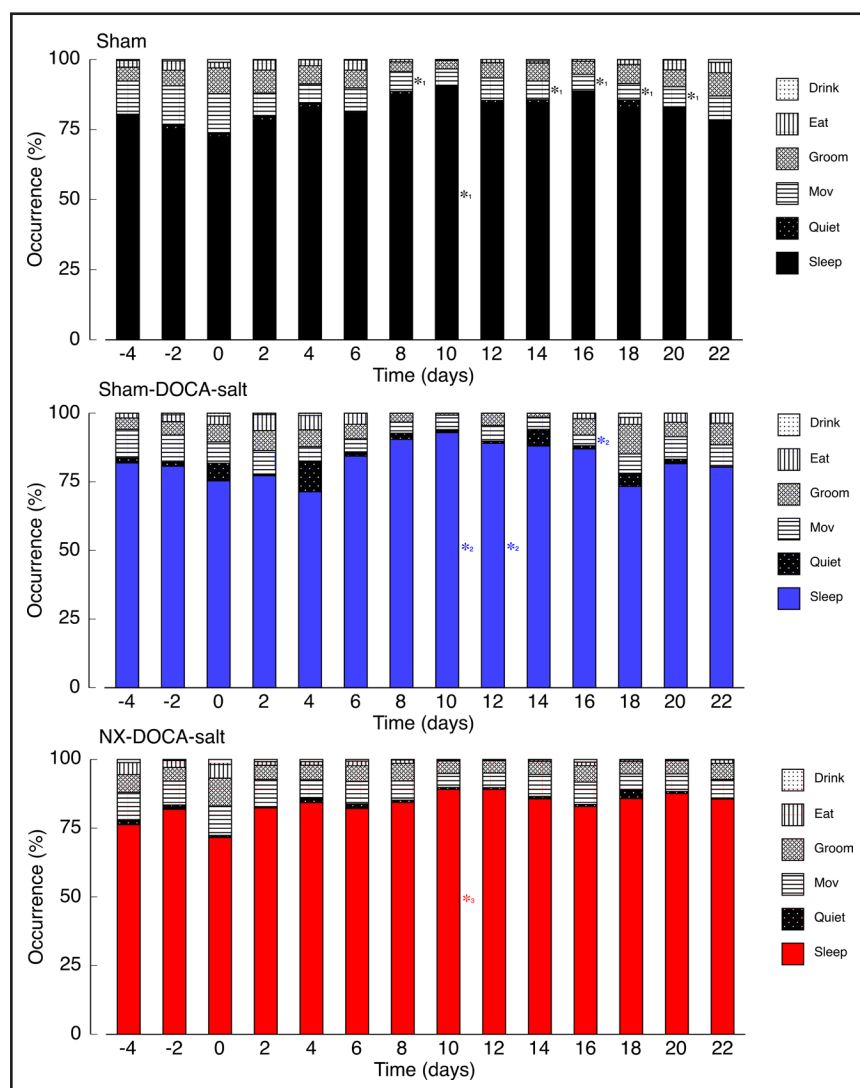


Figure 5. Time course of behavioral state distribution during the light (daytime) phase.

Animal behaviors were categorized into 6 states: sleeping (sleep), quiet awake (quiet), moving (mov), grooming (groom), eating (eat), and drinking (drink). Data are expressed as the percentage of total observation time. Symbols and statistical annotations are defined in the legend of Figure 2. DOCA indicates deoxycorticosterone acetate; and NX, uninephrectomy.

RSNA and LSNA Were Not the Direct Driving Forces Underlying the Initiation and Development of DOCA-Salt-Induced Hypertension

During the initiation phase, AP rose immediately after DOCA-salt administration in both sham and uninephrectomy rats, with significant increases during NREM sleep, which reflects basal levels on day 1 (Figure 2). Concurrently, RSNA and LSNA decreased significantly during NREM sleep on day 1 in both sham-DOCA-salt and uninephrectomy-DOCA-salt rats. These findings indicate that the immediate increase in AP was not directly mediated by RSNA or LSNA.

To the best of our knowledge, no previous study has continuously monitored RSNA and LSNA during the initiation period. Yemane et al⁴ reported in a review that RSNA remained unchanged for 10 days following DOCA-salt administration. Moreover, studies involving total renal denervation,^{5,19} selective afferent renal denervation,¹⁹ or celiac sympathectomy⁹ demonstrated that DOCA treatment during the initiation phase (days 1–5) still produced a significant increase in AP compared with sham animals, indicating that the early hypertensive response persisted despite sympathetic ablation. Collectively, our present findings and previous denervation studies strongly suggest that RSNA, LSNA, and possibly global SNA are not direct contributors to the increase in AP during the initiation phase of DOCA-salt treatment.

Furthermore, the similar AP response observed in uninephrectomy-DOCA-salt and sham-DOCA-salt groups, suggests that unilateral nephrectomy is not the primary factor responsible for the early increase in AP. This is consistent with Wang et al,²⁰ who reported that AP increased in all DOCA-treated groups (sham, uninephrectomy, and spontaneous 1-kidney rats), with greater elevations in the spontaneous 1-kidney and uninephrectomy groups only after 2 weeks of treatment.²⁰ Therefore, both sympathetic activity and unilateral nephrectomy can be excluded as primary determinants of the initial AP rise induced by DOCA-salt treatment.

The developmental phase of DOCA-salt hypertension (days 6–17) was characterized by progressive AP elevation, particularly in uninephrectomy-DOCA-salt rats, beginning on the night of day 6 (Figures 2 and 3A). Notably, at this time, both RSNA and LSNA were significantly suppressed in the uninephrectomy group. These findings indicate that neither RSNA nor LSNA was the primary driving force behind the MAP elevation.

Although no previous study has directly monitored SNA during this phase, Kandlikar et al⁵ reported that renal-denervated rats displayed a similar developmental increase in AP. Plasma norepinephrine concentration, clearance, and spillover did not differ between the sham-DOCA and renal-denervated DOCA groups on days 7 and 14. Additionally, plasma renin activity was markedly

suppressed in DOCA-salt-treated rats.^{21,22} A significant negative correlation between renal norepinephrine content and AP was also observed, indirectly supporting a reduction in RSNA in this model.¹ Taken together, these findings strongly suggest that even during the developmental phase, RSNA, LSNA, and global SNA were not the primary drivers of AP elevation in uninephrectomy-DOCA-salt rats.

Following the cessation of DOCA-salt treatment, both RSNA and LSNA exhibited transient nighttime increases (Figure 2), while AP started to decrease. These responses suggest a reactive sympathetic activation that attempts to sustain the elevated AP but ultimately fails. Thus, the decline in AP during recovery cannot be attributed to reduced RSNA or LSNA.

Taken together, observations across the 3 distinct phases of AP response induced by DOCA-salt treatment provide robust evidence that neither RSNA nor LSNA served as the primary driving force behind the changes. Rather, the initiation and development of DOCA-salt-induced hypertension appear to arise from nonsympathetic mechanisms.

Narrowing the Potential Causal Mechanisms in DOCA-Salt-Induced Hypertension

If SNA is excluded as the direct cause of hypertension, the underlying mechanisms remain uncertain. Although we could not identify the precise factors, the observed changes in RSNA, LSNA, and related parameters helped narrow the range of potential contributors to the initiation and development of DOCA-salt-induced hypertension.

An increase in AP can result from increased cardiac output, elevated total peripheral resistance, or both. May²³ reported that aldosterone-induced hypertension was accompanied by increased vascular resistance without significant changes in cardiac output.²³ Our findings are consistent with this interpretation. DOCA-salt administration did not significantly alter water balance or body weight gain (Figure 4), in line with Jacob et al⁶. These results do not demonstrate extracellular fluid expansion and suggest that, if present, any such expansion is unlikely to be substantial. Moreover, RSNA and LSNA did not increase, making enhanced venous return and cardiac preload unlikely, whereas reduced SNA (Figures 2 and 3A) argues against a contributory role of increased myocardial contractility. Taken together, these results indicate that the increase in AP is not attributable to greater cardiac output but rather to an immediate and progressive increase in total peripheral vascular resistance.

Our direct RSNA and LSNA recordings also demonstrated that this increased vascular resistance was not driven by SNA. Consistently, May²³ showed that progressive increases in total peripheral resistance in aldosterone-induced hypertension were not prevented by α - or

β -adrenergic blockade. Instead, previous studies suggest nonsympathetic mechanisms. DOCA-salt hypertension is characterized by sustained vasoconstriction, impaired vasodilatory capacity, and endothelial dysfunction.^{24–27}

Importantly, in the rat DOCA model, neither DOCA nor high salt intake alone induces hypertension.²⁸ Rather, the interaction between DOCA and sodium loading is required for the development of the hypertensive phenotype. Excess dietary sodium has been reported to accumulate in an osmotically inactive form within the interstitial compartments of the skin and skeletal muscle in both humans^{29,30} and experimental animals.³¹ Such interstitial sodium accumulation activates macrophage-mediated immune responses, particularly in perivascular tissues.^{32,33} In parallel, DOCA binds to EC-MRs (endothelial cell mineralocorticoid receptors) and promotes vascular inflammation.^{25,34} Although these pathways are distinct, they appear to converge on shared immune-inflammatory mechanisms. Sodium accumulation within perivascular interstitial spaces and DOCA-dependent EC-MR activation may interact synergistically, leading to increased oxidative stress, enhanced inflammatory signaling, uncoupling of endothelial nitric oxide synthase, and impaired nitric oxide bioavailability.^{34–36} The net effect is vascular dysfunction characterized by attenuated vasodilatory capacity. These vascular effects may be further modulated by renal-derived immune mechanisms^{37,38} or afferent signaling pathways¹⁹ following uninephrectomy. This immune-inflammatory framework is consistent with the rapid, sustained, and reversible modulation of vascular function discussed above, and is therefore more compatible with predominantly nonsympathetic mechanisms underlying the increase in peripheral vascular resistance in the DOCA-salt model.

In summary, the immediate, sustained, and reversible course of AP changes in DOCA-salt hypertension is most likely attributable to an increase in total peripheral vascular resistance arising from nonsympathetic mechanisms, potentially involving endothelial dysfunction and perivascular immune-inflammatory processes under the combined influence of DOCA and salt.

Limitations

Some limitations of this study should be acknowledged. First, although emerging evidence highlights the importance of afferent RSNA,^{19,39,40} assessing afferent activity based on multifiber recordings in conscious rats remains technically challenging. The RSNA recorded here represents the combined activity of multiple sympathetic axons, including both afferent and efferent fibers. In normal rats, multifiber RSNA predominantly reflects efferent activity, as sympathetic ganglionic blockade produces near-complete suppression.⁴¹ Moreover, the amplitude of afferent sympathetic signals, even when pharmacologically maximized, is typically <5 μ V.^{19,42} This signal is only

slightly above twice the noise threshold and can only be reliably detected under stable anesthetized conditions. Given these methodological constraints, even if afferent RSNA were activated by DOCA-salt administration, our approach lacked the sensitivity to detect such changes in conscious animals. Further methodological advances are required to enable reliable evaluation of afferent RSNA under conscious conditions.

Second, we cannot exclude the possibility of excessive SNA activation during later developmental and maintenance phases of DOCA-salt hypertension. The present study focused on RSNA and LSNA changes during the initiation and early developmental phases; therefore, our conclusions cannot be generalized to all phases of the disease. We observed that RSNA tended to recover on days 15 to 17 in the uninephrectomy-DOCA-salt group, approaching baseline levels (Figure 3A). Plasma norepinephrine concentrations have been reported to increase at 4 and 7 weeks of DOCA-salt hypertension,⁴³ whereas both norepinephrine levels and spillover remain unchanged at 1 to 2 weeks.⁵ Thus, it remains possible that heightened SNA contributes to hypertension during the late developmental and maintenance phases.

Third, an interesting finding of the present study is that changes in HR did not consistently parallel changes in renal or lumbar SNA. However, SNA is regulated in a region-specific manner⁴¹ and is influenced by vagal modulation and baroreflex mechanisms. In particular, in chronic or time-dependent conditions such as hypertension, HR may not fully reflect sympathetic nerve outflow. The absence of a detailed discussion of this dissociation based on the present results represents a limitation of the study.

Finally, our findings highlight the potential importance of sodium accumulation and its dynamics in the initiation of DOCA-salt hypertension. A substantial body of previous work has demonstrated that sodium balance becomes significantly positive following DOCA administration and continues to increase cumulatively (#), and that total body sodium content increases markedly, with most retained sodium accumulating in interstitial compartments such as the skin without proportional water retention (#). However, sodium balance was not directly measured in the present study, and our discussion of this aspect relies on findings reported in the existing literature. We acknowledge this as a limitation of the present study design.

Perspective

We previously conducted continuous recordings of RSNA and LSNA in hypertensive rat models, including Ang II + salt-induced hypertension¹⁰ and Dahl salt-sensitive rats,¹¹ throughout the transition from normotension to hypertension. In the present study, we extended this approach to the DOCA-salt hypertensive rat model. Combining these

findings, these studies reveal a consistent pattern: RSNA decreases, while LSNA remains unchanged or decreases in salt-induced hypertension models in which an additional factor, Ang II (angiotensin II), mineralocorticoids, or genetic predisposition, is present.

In all 3 models examined, neither RSNA nor LSNA increased. This indicates that heightened SNA is not a causative factor in the initiation and development of hypertension. Although excessive salt intake in combination with Ang II, mineralocorticoids, or genetic factors is required for hypertension onset, our long-term recordings suggest that hypertension is not directly mediated by osmotically induced increases in sympathetic outflow. Instead, nonsympathetic mechanisms appear to underlie the progressive elevation of AP in these models.

We propose that salt-induced hypertension arises from the interaction of high salt intake with factors such as DOCA, Ang II, or genetic predisposition, which collectively trigger immune-inflammatory responses that impair vascular function and elevate peripheral resistance. These findings highlight immune-inflammatory pathways as potential therapeutic targets in salt-induced hypertension.³⁵

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Disclosures

None.

Supplemental Material

Supplemental Methods

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